CM 2020 Agile Software Projects

Midterm Project - Proposal

SAGE - A Mental Health Companion App

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Introduction

Background

In recent years, the prevalence of Major Depressive Disorder (MDD) and Generalised Anxiety Disorder (GAD) has become a significant concern globally. According to the World Health Organization (WHO), approximately 280 million people worldwide suffer from depression, with about 5% of adults affected by this mental disorder. Depression is particularly prevalent among women, who are 50% more likely to experience it compared to men. Additionally, more than 700,000 people die due to suicide each year, making it the fourth leading cause of death among individuals aged 15 to 29. (*Depressive Disorder (Depression)*, 2023)

In Singapore, mental health issues such as depression and anxiety are increasing as well. According to a recent survey, about 25.3% of young adults in Singapore report experiencing poor mental health. This trend is particularly concerning among younger demographics, representing the highest proportion of affected individuals (Ganesan, 2023). Additionally, studies have shown that approximately 14% of Singaporeans experience symptoms consistent with depression, while 15% report symptoms of anxiety. ("IMH: Cost of Anxiety and Depression in Singapore Runs Into the Billions," 2023).

Given this backdrop, mental health companion apps have emerged as valuable tools to support individuals dealing with these conditions. These apps typically offer features such as mood tracking, therapeutic exercises, access to informational resources, and community support, aiming to provide comprehensive assistance to users struggling with depression and anxiety.

Problem Statement

Despite the widespread availability of mental health apps, several challenges persist. Many of these apps struggle to provide personalised and empathetic responses tailored to individual user needs, relying heavily on automated scripts rather than offering genuine emotional engagement. Additionally, while some apps excel in content delivery, they often fail to foster real-time, interactive companionship crucial for immediate emotional support.

Aims & Objectives

Objective

To introduce a novel mental health companion app specifically tailored to the diverse needs of young adults in Singapore aged 18 to 29, addressing the current limitations in personalisation, empathy, and real-time interaction.

Aims

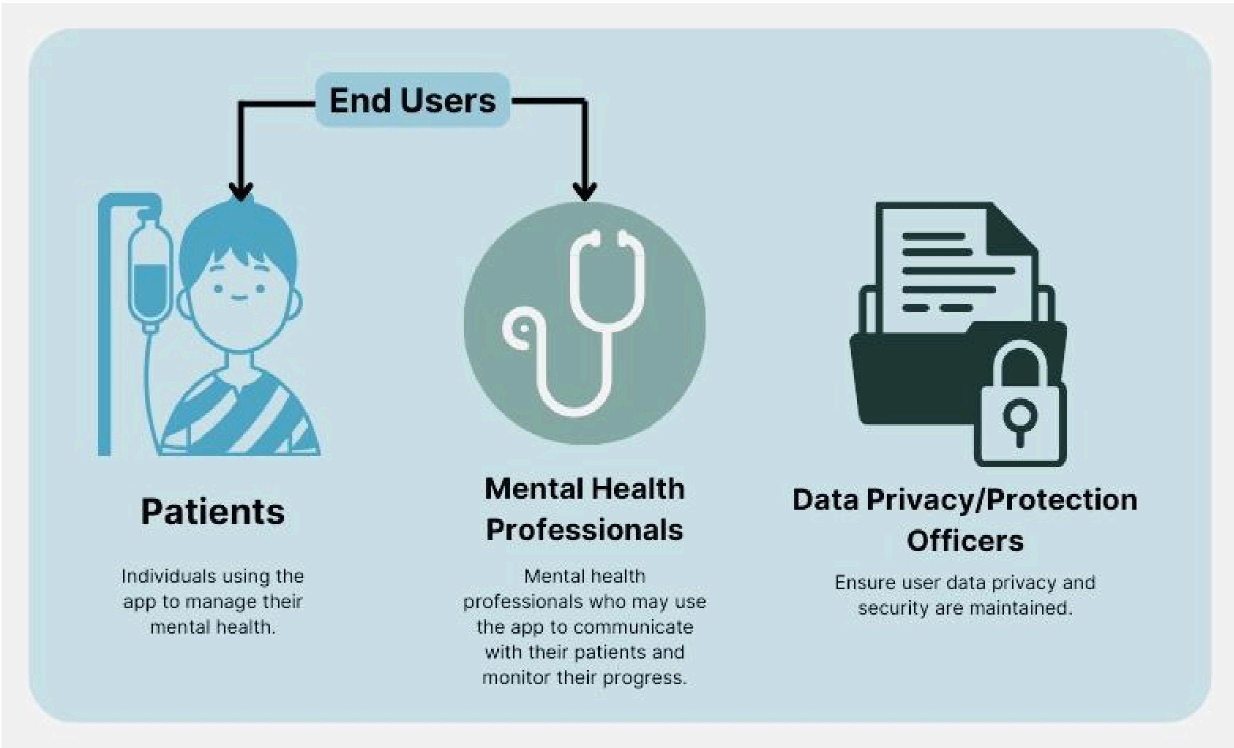
Our approach focuses on delivering comprehensive tools to enhance mental well-being. We prioritise implementing robust mood management tools, including intuitive features for mood tracking and progress monitoring, empowering users to gain insights into their emotional health over time. Central to our vision is redefining digital companionship by integrating advanced AI technology and empathetic design principles. This innovative approach ensures that our app offers more than just functionalities—it provides users with real-time guidance and empathetic responses tailored to their emotional needs, fostering a sense of trust and companionship. Ultimately, we aim to create a supportive environment where individuals can manage their mental health effectively, accessing resources and support that promote healing and resilience.

Expected Impact

Our ultimate goal is to create an innovative mental health companion app that effectively reaches and profoundly benefits its intended audience, fostering a safer, more supportive environment for mental health management.

Requirements

Stakeholders



For our mental health companion app, we have identified **three key stakeholders** essential to its success: Patients (users), mental health professionals, and data protection officers.

Patients are at the heart of our app's design, and understanding their needs and preferences is crucial for creating a supportive and effective platform. By prioritising their feedback and incorporating features like mood tracking, guided meditations, and secure communication options, we aim to provide a personalised and empowering experience for individuals navigating depression and anxiety.

Mental health professionals represent another critical stakeholder group. They may use our app to augment patient care through secure communication and resource sharing. Their insights and collaboration ensure that our app meets clinical standards and enhances therapeutic outcomes.

Lastly, engaging **data protection officers** ensures rigorous compliance with privacy regulations and industry standards, safeguarding user data through secure storage practices and transparent data handling policies. Together, these stakeholders form a collaborative network dedicated to fostering mental well-being while upholding the highest standards of user privacy and security.

Possibly in the future:

- **Technology Partners:**
    - API Providers: Companies providing essential services like authentication, payment processing, or third-party integrations.
    - Cloud Service Providers: Ensuring reliable hosting and data storage solutions.
- **Healthcare Regulators**: Organisations ensuring the app complies with healthcare regulations (i.e. MOH Singapore)
- **Healthcare Providers:**
    - Hospitals and Clinics: Institutions that may recommend or integrate the app into their services.
    - Mental Health Organisations: Organisations focused on mental health advocacy and support.

User Research and Needs Analysis

To ensure the app meets the needs and expectations of our target users, we will conduct comprehensive surveys and in-depth interviews to understand their needs, pain points, and preferences. The results will be meticulously analysed to define the problem statement and key requirements, guiding us in prioritising a prototype's most impactful features and specifications.

Subsequently, we will present the prototype for user testing, gathering feedback through additional surveys and interviews to refine our solution statement. This iterative process will provide a clear direction for the app's future development, ensuring it evolves to effectively meet user needs.

Interview Data Analysis[1]

As of **26 June 2024**, we have completed **5 of the 30** intended interviews we will conduct in this project. Even though we were hoping to gain more insight from professionals in the mental health industry, we could not find one available for an interview. Hopefully, by the middle of the

[1] Please refer to Appendix A for Interview Questionnaire.

project, we will find at least three professionals with their insight. The following analysis and insights have been made according to the data we have obtained from the interviews:

Privacy and User Interface

- All subjects expressed **high concern about the privacy and security of their data** when using mental health apps and were cautious about apps asking for too much personal information to provide personalised recommendations.
    - Subjects preferred measures include:
        - clear T&Cs
        - data storage on personal devices
        - reputable cloud providers
        - data deletion options.
- A preference for **clear, simple**, and **minimalistic** interfaces, with easy navigation and minimal complexity highlighted as important features.
- Subjects appreciated apps that were **aesthetically pleasing, user-friendly**, and offered **progress tracking.**
- While specific accessibility needs were not highlighted by most subjects, **ensuring readability** and **easy-to-use features** was mentioned

App Content & Features

- While **personalised** content and recommendations were generally seen as helpful, they should be **optional** and **non-intrusive.**
- **Guided meditations** and **mood tracking** were consistently mentioned as valuable features.
- Providing **educational content and resources on mental health** was considered important for reducing stigma and offering practical help, with credible articles and professional resources preferred.
- **Therapist communication** was also considered important by multiple subjects
- Some valued **community support** or a **sense of community**, although one subject preferred professional support over community interaction

User Engagement

- **Effective solutions and practical insights** were seen as key motivators for regular app use.

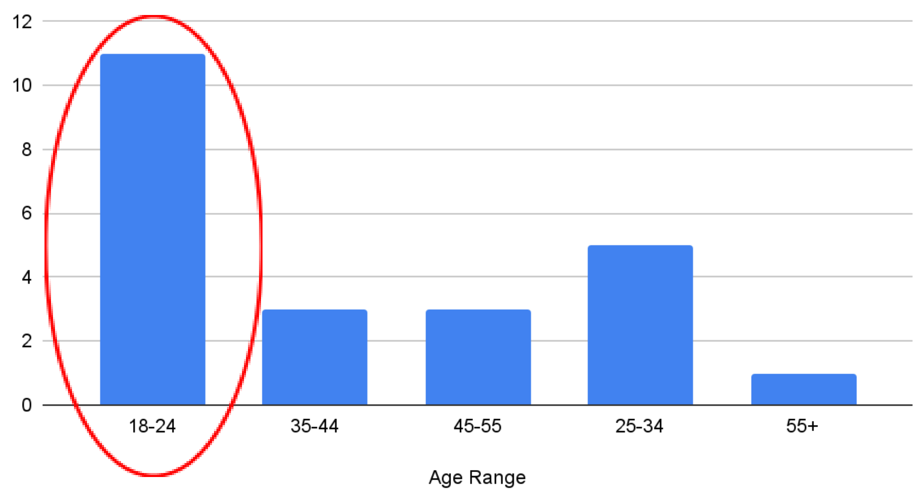
- **Reward systems and gamification** were mentioned, but their effectiveness varied among individuals.
- **Notifications should be customisable** to avoid fatigue, with a preference for relevant and limited notifications.
- **Common dislikes** included paywalls, repetitive content, generic responses from chatbots, and a lack of in-depth solutions.

Survey Data Analysis[2]

As of *Saturday, **15 June 2024***, there have been ***23* responses from our survey.**
Upon extracting the raw data, we proceeded to clean and prepare the data by removing unnecessary and redundant data and converting scale responses into numerical format (1–5).

Demographic Analysis Findings

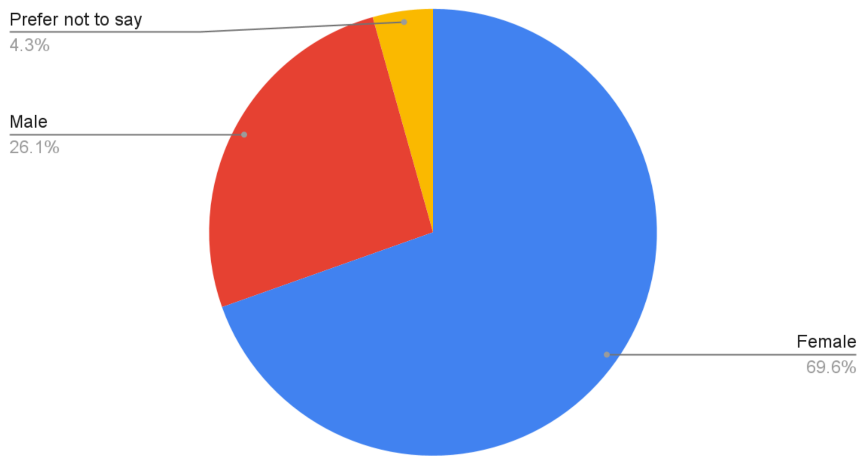
Age Distribution



- Most respondents fall into the **18-24 age group,** followed by **25-34, 35-44,** and **45-55 age groups.**

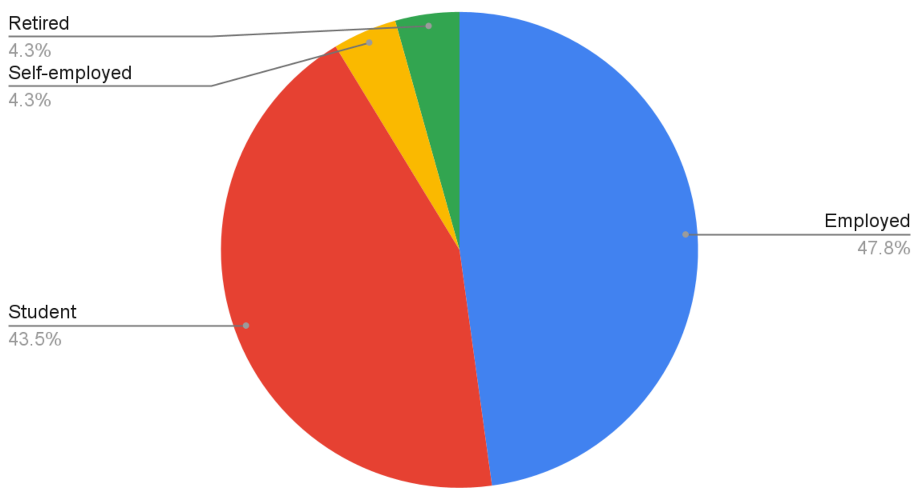
[2] Please refer to Appendix C for Survey Form.

Gender Distribution



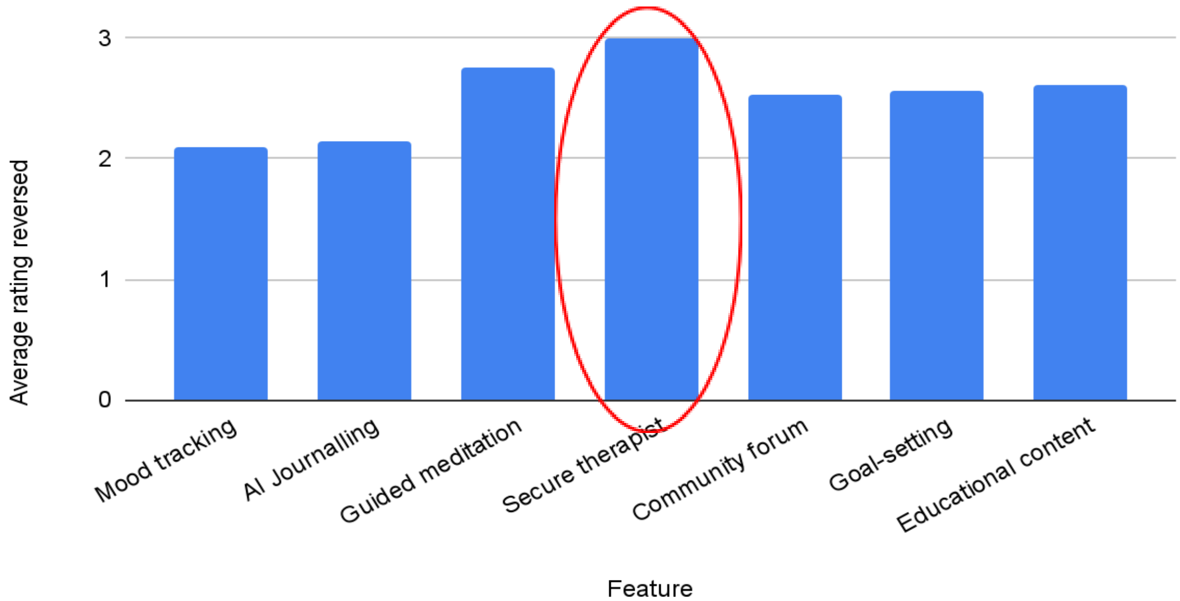
- The majority of respondents are **female.**

Employment Status



- Most respondents are **employed,** followed by **students.**

Functional Features Findings



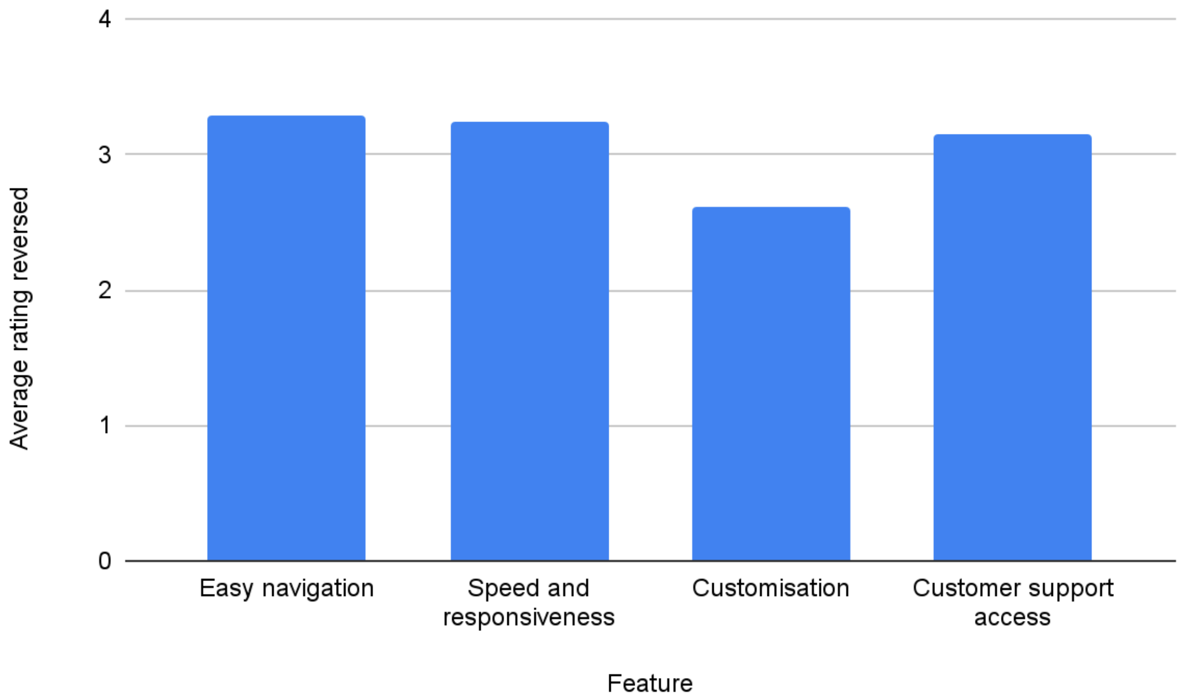
The surveyees were told to rate seven functional features of the app according to their importance (i.e., 5 = very important/useful; 1 = Not important/useful).:

- Mood Tracking
- Journalling
- Meditation
- Therapist Communication
- Community group support (forum)
- Goal Setting
- Educational Content

The following are the findings collected:

- **High importance** for **therapist communication.**
- Moderate likelihood of using educational content and guided meditation features.
- Moderate interest in goal setting and community forums.

Non-functional Features Findings



The surveyees were also asked to rank the importance of four non-functional features.
(i.e., 5 = very important/useful; 1 = Not important/useful).

- Easy navigation
- Speed and responsiveness
- Customisation
- Customer support access

The following were the findings gathered:

- **High importance** for **Easy navigation,** and **Speed and responsiveness.**
- Moderate importance for Customisation and Support availability

Additional Feedback Findings

Surveyees were also asked to give additional feedback based on what they would like to see in their ideal mental health companion app:

Suggestions include:

- Suggestion for a feature to store favourite learnings from self-help books.
- Suggestion to incorporate a video diary feature.
- Importance of tracking food and exercise, especially during mental health spirals.
- Suggestion to add fun features like mind games to engage users.
- Proposal for an SOS feature to assist users in deep distress.
- Suggestion for a shop selling journals, worry stones, and other mental health aids, including an online pharmacy for prescription drugs.
- Emphasis on making the app user-friendly across different age groups.
- Important to make the features easy and convenient to use.
- Concerns about data privacy and secure storage.

Conclusion from Evidence

From all the findings collated through the survey and interviews, we have managed to draw out several user stories and categorise them according to the potential features they have ranked previously in the survey.[3]

Based on the interview and survey analysis and findings, we have highlighted the following points about the features:

1. **Focus on Key Features:** Prioritise the development of mood tracking, goal setting, and educational resources with guided meditations, chatbot and therapist communication.
2. **Consider Additional Features:** Explore the feasibility of adding features like a vlog diary, food and exercise tracking, and mind games. Suggestions for user engagement, such as gamification and a reward system, were also discussed.
3. **Enhance Privacy and Security:** Implement robust data privacy and storage measures to address user concerns.
4. **User Experience:** Ensure the app is user-friendly, simple and accessible to a wide range of users, including different age groups.

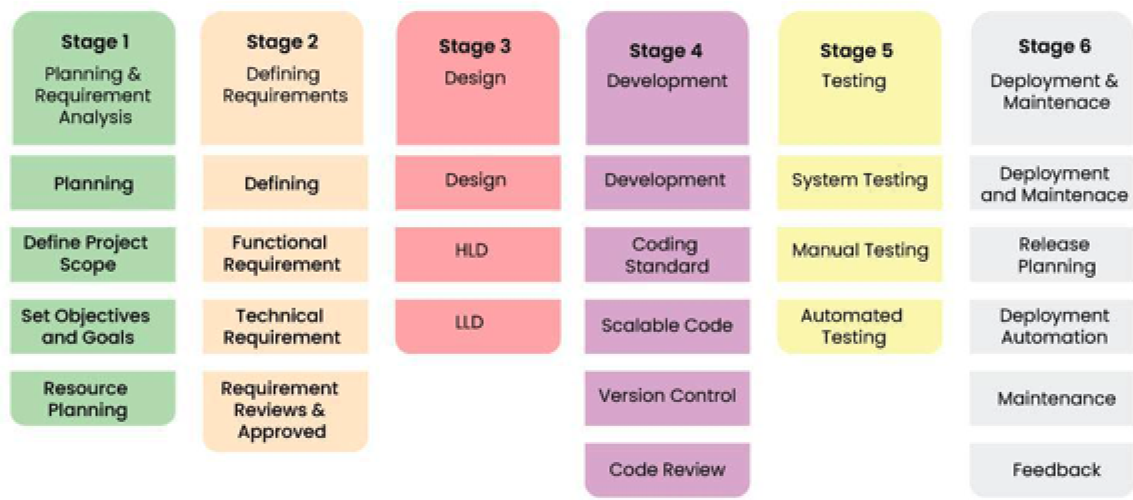
[3] Please refer to Appendix B for the User Stories

Planning

Our team will follow the phases listed in the ***Software Development Lifecycle****,* as shown below, together with the **SCRUM** framework for project management:

Software Development Life Cycle

The SDLC framework guides our multi-phase process, which begins with comprehensive user research and needs analysis, followed by detailed development planning and execution. The ultimate goal is to deliver a user-centred, highly functional, and supportive tool for managing mental health.



(Source: https://www.geeksforgeeks.org/software-development-life-cycle-sdlc)

Main Tasks

Stage		Backlog Tasks
1	Project Prep	Set up the SCRUM framework.
2	Requirement Analysis & Feasibility Study	Conduct initial interviews for the problem statement with 25 more people.
		Start gathering user stories and prototyping core features.
3	Architectural Design	Create wireframes for the user-friendly mood-logging interface.

		Design visualisations for mood trend tracking.
		Design motivational content and reminders.
		Curate guided meditations specifically for depression and anxiety symptoms.
4	**Software Development**	Develop UI components for mental health goal setting and progress tracking.
		Implement the journaling feature with AI-generated prompts and Chatbot.
		Integrate audio and video content API
5	**Testing**	Conduct usability testing sessions with 30 target users.
		Gather feedback on prototypes.
6	**Review**	Iterate designs to improve usability and functionality and address identified issues.
7	**Deployment**	Prepare for the beta launch.
		Release a beta version to a selected user group.
		Collect and analyse user feedback, and identify areas for improvement and additional features.

Stages 3 to 7 will be repeated with every new release of a feature or further enhancements of the app.

SCRUM

Release and Sprints

The following table[4] depicts the estimated timeline and dependencies for the main tasks identified thus far. We have structured our timeline to achieve a beta release, projected to be completed **between July 1st and November 18th**, with the beta release scheduled for

[4] Please refer to Appendix B, User Stories for a more detailed user story of each feature/theme.

November 19th, 2024. It is important to note that by the final submission of this school project, we will have yet to reach the first beta release of the app.

Release	Sprint (2 weeks)	Epic (Feature/CustomerRequest/BizRequirement)	Theme
Final Project Submission 8th Sept 2024	Sprint 1	Develop a user-friendly interface for logging daily moods.	Mood Tracking
		Implement visualisation tools for users to track mood trends over time.	
		AI will analyse data from mood tracking for trends and make a “forecast” of mood.	
	Sprint 2	Create a journaling feature that provides AI-generated open-ended prompts to encourage reflective writing.	Journaling
		Store users' journal data in cloud storage	
	Sprint 3	AI GPT bot (configured for mental health awareness like Replika)	Crisis Management
	Sprint 4	Implement tools for users to set personal mental health goals and track their progress.	Goal Setting and Progress Tracking
		Provide motivational content and reminders to help users stay on track.	
	Sprint 5	Curate a library of guided meditations specifically designed to alleviate symptoms of depression and anxiety.	Guided Meditations
		AI to recommend meditations and articles based on mood.	
Beta Release 18th Nov 2024	Sprint 6	Create a community forum	Community Support
	Sprint 7	Develop a secure messaging feature for users to communicate with their therapists.	Secure Support
		AI to recommend meditations and articles based on mood	Mental Health Education

	Sprint 8	Create a journaling feature that provides AI-generated open-ended prompts to encourage reflective writing.	Personalised Content
		Ensure the AI is adaptive and responsive to user inputs, offering relevant and supportive prompts.	
	Sprint 9	Create options for security and features in settings	Customisation
		Create themes and layout options in settings.	
	Sprint 10	Create reminder tool	Reminders and Notifications
		Create notification tool	

Each sprint is performed with a rigorous structure, as shown below. This ensures that each sprint is effectively managed and that all critical aspects of development and testing are addressed promptly. Here's Sprint 1, for example -

Sprint 1 Goal: Finalise Mood Tracking Features

Each Sprint Duration: 2 weeks (14 days)

Sprint Timeline:

Day 1-3: Start initial coding

- Planning workshop
- Beginning sketching out front-end and back-end coding.
- Ensure all features meet the predefined acceptance criteria.

Day 4-6: Testing and Debugging

- Conduct extensive unit testing, integration testing, and system testing.
- Identify and fix critical bugs to ensure stability.
- Perform user acceptance testing (UAT) with a small group of target users to gather initial feedback.

Day 7-9: Quality Assurance and Refinement

- Conduct a thorough quality assurance (QA) review, focusing on usability, performance, and security.
- Refine UI/UX based on feedback from UAT and QA findings.

- Implement any minor enhancements that improve user experience without introducing new risks.

Day 10-11: Documentation and Preparation

- Finalise user documentation, including help guides and FAQs.
- Prepare release notes detailing the features, improvements, and known issues.
- Set up deployment pipelines and ensure all release processes are in place.

Day 12-13: Internal Testing Deployment

- Deploy the internally tested version of the app to a selected group of internal testers.
- Monitor the deployment process to address any issues that arise quickly.
- Ensure communication channels are open for immediate feedback from internal testers.

Day 14: Post-Testing Monitoring and Feedback Collection

- Monitor the app's performance and user feedback.
- Collect data on app usage, user satisfaction, and any reported issues.
- Begin planning for the next sprint based on the feedback and performance metrics.

The remaining sprints will be planned out in advance after the midterm proposal submission, and we will follow them for the rest of the project. Our final proposal for this module's end submission will feature a detailed look at the remaining sprints.

Contingencies

Considering the tasks we have allocated and defined, there are several contingencies that we may have to consider:

1. There may be resource constraints, especially with time, as we only have 9 weeks before our final project submission deadline; some sprints further along the timeline may be replanned where some app features will be prioritised for completion first so that we will at least have an app with minimum viable functionality to present during the final project submission.

2. Team members are unavailable due to other personal commitments. To be prepared for situations like this, we always ensure that at least two team members handle each task in case one has to take leave.

3. We may also encounter unforeseen technical issues during development, such as integration problems with APIs or unexpected bugs. As such, we will consult our lecturer and expert colleagues for help and promptly address unforeseen issues during development. Buffer time will be allocated to the project timeline for technical issues.

4. If we are to do beta testing and If beta feedback takes longer than expected, we will plan for additional iterations.

Meeting Notes

During our weekly meetings while coming up with the concept and discussion of this proposal, we have done up some meeting notes for the sessions. The raw meeting notes can be referred to in Appendix E.[5]

[5] Please refer to Appendix E for the raw meeting notes.

Scope

Geographical Scope

- **Within Singapore Only:**
  - The app is designed specifically for users residing in Singapore. This geographic limitation allows for tailored content, resources, and support services relevant to the local context. This includes integrating local mental health organisations, resources available in Singapore, and content in languages spoken in Singapore.

Content and Feature Limitations

- **Specific Mental Health Conditions**:

  The initial version of the app primarily focuses on depression and anxiety. While these are common and impactful conditions, other mental health issues such as bipolar disorder, PTSD, and eating disorders may not be fully addressed in this version.

- **Therapist Availability:**

  The secure communication feature depends on therapists' availability and willingness to use the platform. Users may need to have their therapists or be connected through partnered organisations.

- **Providing Conclusive Effectiveness**

  Assessing the effectiveness of our assistance is challenging due to the nuanced nature of determining what constitutes effective support. Quantifying the impact or efficacy of specific features is inherently complex and difficult to measure accurately. As a result, some users may not perceive the app as effective.

Technical and Resource Constraints

- **Scalability:**
  - The initial beta release may have limited scalability, affecting performance as the user base grows. Future updates will focus on improving scalability and performance.

- **Development and Maintenance:**
    - The project's scope includes ongoing development and maintenance, but resource constraints may limit the speed and extent of feature rollouts and updates.
- **Time constraints:**
    - Only be able to input the AI feature of mood tracking, chatbot and journaling, can't expand on the forum yet as we need human moderators as well on top of the moderator bot
- **Professional input:**
    - No input from psychology professionals at the time of proposal submission

Specification

Features

Feature	Details
1. Mood Tracking	Develop a user-friendly interface for logging daily moods.
	Implement visualisation tools for users to track mood trends over time.
	AI will analyse data from mood tracking for trends and make a "forecast" of mood.
2. Goal setting and Progress Tracking	Implement tools for users to set personal mental health goals and track their progress.
	Provide motivational content and reminders to help users stay on track
3. Journaling with AI Prompts	Create a journaling feature that provides AI-generated open-ended prompts to encourage reflective writing.
	Ensure the AI is adaptive and responsive to user inputs, offering relevant and supportive prompts.
	Data is also used to supplement the AI for mood trend analysis.
4. Guided Meditations and Useful Articles	Curate a library of guided meditations specifically designed to alleviate symptoms of depression and anxiety.
	Integrate audio and video content seamlessly within the app.
	AI to recommend meditations and articles based on mood trends.
5. Chatbot	AI GPT bot (configured for mental health awareness like Replika)
6. Secure Therapist Sessions	Develop a secure message feature for users to communicate with their therapists.
	Implement encryption and data protection measures to ensure privacy.
7. Community Forum	Moderated forums (by both bot and human) for users to mingle and share their journey of struggles and recoveries.

Functionality Requirements

1. Mood Tracking

User Interface (UI) Design:

- **Wireframes**
  - Initial sketches and wireframes to outline the mood logging interface.
  - Create low-fidelity wireframes to map out user interaction and data entry flow.
  - Develop high-fidelity prototypes with detailed UI elements and visual design.
  - Example Tools: Figma, Sketch, Adobe XD.
- **Components**
  - **Mood Selection:** Buttons or icons representing moods (e.g., happy, sad, anxious).
  - **Notes Section:** Text input fields for users to add context to their mood entries.
  - **Rating Scale:** Slider or star-rating component to rate mood intensity.
  - **Calendar View:** Calendar component to display mood entries over time.
  - **Graphical Visualisation:** Line charts, bar charts, and pie charts to show mood trends.

Coding and Technical Implementation:

- **Frontend Development**
  - Technologies: HTML, CSS, JavaScript, React or Angular.
  - Implement responsive design for different devices (mobile, tablet, desktop).
  - Utilise libraries like Chart.js or D3.js for data visualisation.
- **Backend Development**
  - Technologies: Node.js, Express, Python (Django or Flask).
  - Design API endpoints for creating, reading, updating, and deleting mood entries (CRUD operations).
  - Ensure secure data transmission and storage (use HTTPS, JWT for authentication).
- **Database Design**
  - Use SQL (PostgreSQL) or NoSQL (MongoDB) databases.
  - Schema Example: **`users`** table, **`moods`** table with fields for date, mood type, intensity, notes, user ID.

AI Analysis:

- **Machine Learning Models**
  - Use libraries like TensorFlow or PyTorch.
  - Develop models to analyse mood data and forecast trends.
  - Train models using historical mood data to identify patterns.
- **Integration**
  - Implement API endpoints to retrieve AI analysis results.

    - Display forecasts and insights on the user dashboard.

2. Goal Setting and Progress Tracking

UI Design:

- **Wireframes**
    - Design interfaces for setting and tracking personal mental health goals.
    - Create interactive elements for adding, editing, and completing goals.
    - Visual progress tracking with charts and progress bars.
- **Components**
    - **Goal Creation Form:** Input fields for goal title, description, and target date.
    - **Progress Bars:** Visual indicators of goal completion percentage.
    - **Motivational Content:** Sections for reminders and inspirational messages.

Coding and Technical Implementation:

- **Frontend Development**
    - Technologies: React Native (for mobile), React/Angular (for web).
    - Implement interactive forms and progress indicators.
    - Use animations for smooth user interactions.
- **Backend Development**
    - Design API endpoints for goal management (CRUD operations).
    - Implement notification service for reminders using Firebase Cloud Messaging or similar.
- **Database Design**
    - Schema Example: **`goals`** table with fields for goal ID, user ID, title, description, target date, completion status.

3. Journaling with AI Prompts

UI Design:

- **Wireframes**
    - Layout for a text editor with AI-generated prompts.
    - Adaptive prompt suggestions based on user inputs.
    - Data integration for mood analysis and trend visualisation.
- **Components**
    - **Text Editor:** Rich text editor with formatting options.
    - **Prompt Display:** Dynamic section for AI-generated prompts.
    - **Submission Button:** For saving journal entries.

Coding and Technical Implementation:

- **Frontend Development**
    - Libraries: Draft.js, Quill.js for rich text editor functionality.

    - Integrate real-time AI prompts using WebSockets or similar.
- **Backend Development**
    - Implement AI algorithms using NLP libraries like spaCy or Hugging Face Transformers.
    - API endpoints for saving and retrieving journal entries.
- **Database Design**
    - Schema Example: **`journal_entries`** table with fields for entry ID, user ID, text content, prompt used, date.

4. Guided Meditations and Useful Articles

UI Design:

- **Wireframes**
    - Design a library interface for guided meditations and articles.
    - Integrate audio and video content seamlessly.
    - Personalisation through AI recommendations.
- **Components**
    - **Library Interface:** Grid or list view for browsing content.
    - **Playback Controls:** Play, pause, skip, and download options for audio/video content.
    - **Recommendation Section:** Personalised content suggestions based on user data.

Coding and Technical Implementation:

- **Frontend Development**
    - Technologies: React Native for mobile, React/Angular for web.
    - Integrate media players using HTML5 video/audio or libraries like Video.js.
- **Backend Development**
    - APIs for fetching, streaming, and downloading content.
    - AI recommendation algorithms using collaborative filtering or content-based filtering.
- **Database Design**
    - Schema Example: **`meditations`** and **`articles`** tables with metadata fields (title, author, duration, tags).

5. Chatbot

UI Design:

- **Wireframes**
    - Chat interface layout with real-time messaging.
    - Bot responses and user input sections.
    - Escalation protocols for emergencies.

- **Components**
    - **Chat Interface:** Message bubbles for user and bot interactions.
    - **Input Field:** Text input for user messages.
    - **Emergency Protocols:** Buttons for contacting support or emergency services.

Coding and Technical Implementation:

- **Frontend Development**
    - Libraries: Botpress, Rasa for chatbot implementation.
    - WebSockets for real-time messaging.
- **Backend Development**
    - NLP models using GPT-4 for generating responses.
    - Integration with backend services for user data and mood analysis.
- **Database Design**
    - Schema Example: **`chat_logs`** table with fields for user ID, message content, timestamp.

6. Secure Therapist Sessions

UI Design:

- **Wireframes**
    - Secure messaging interface with encryption.
    - Scheduling and document sharing features.
    - Privacy controls for users.
- **Components**
    - **Messaging Interface:** Encrypted chat with message bubbles.
    - **Scheduler:** Calendar for booking sessions.
    - **Document Sharing:** Upload and download functionalities.

Coding and Technical Implementation:

- **Frontend Development**
    - Technologies: React Native for mobile, React/Angular for web.
    - Implement secure messaging using WebSockets and SSL/TLS.
- **Backend Development**
    - Encryption protocols (e.g., RSA, AES) for secure communication.
    - APIs for managing appointments and document sharing.
- **Database Design**
    - Schema Example: **`sessions`** table with fields for session ID, user ID, therapist ID, scheduled time.

7. Community Forum

UI Design:

- **Wireframes**
  - Forum layout with categories and threads.
  - Moderation tools for ensuring a safe environment.
  - User interaction features like posting, commenting, and reacting.
- **Components**
  - **Forum Categories:** Sections for different topics.
  - **Thread Interface:** List of threads with sorting and filtering options.
  - **Post Interaction:** Options to like, comment, and report posts.

Coding and Technical Implementation:

- **Frontend Development**
  - Technologies: React Native for mobile, React/Angular for web.
  - Implement real-time updates using WebSockets.
- **Backend Development**
  - Moderation tools using AI for detecting harmful content.
  - APIs for forum management (CRUD operations for posts, comments).
- **Database Design**
  - Schema Example: **`forums`**, **`threads`**, and **`posts`** tables with relational fields for user interactions.

Non-Functional System Requirements

1. Performance Requirements

- **Response Time:** The app should respond to user actions within 2 seconds for 95% of transactions to ensure a seamless user experience when logging moods, interacting with the chatbot, or accessing guided meditations.
- **Scalability:** The app must handle up to 10,000 concurrent users without degradation in performance, ensuring it can support a growing user base seeking mental health support.
- **Throughput:** During peak times, such as mental health awareness campaigns or peak hours, the system should process at least 500 transactions per second.
- **Latency:** Network latency should not exceed 200 milliseconds for any action within the app, including mood tracking and journaling.

2. Reliability Requirements

- **Uptime:** The app should have an uptime of 99.9%, ensuring it is available 24/7 for users who might need immediate mental health support at any time.
- **Error Rate:** The system should have an error rate of less than 0.1% for all transactions, ensuring reliability in mood tracking, journaling, and accessing resources.

- **Data Consistency:** Ensure data consistency across all modules, particularly mood logs, journal entries, and user profiles, with regular integrity checks.
- **Backup and Recovery:** Daily backups of all user data, including mood logs and journal entries, must be performed, with a recovery point objective (RPO) of 1 hour and a recovery time objective (RTO) of 2 hours to prevent data loss.

3. Security Requirements

- **Data Encryption:** All data in transit and at rest must be encrypted using AES-256 to protect sensitive user information. (Rijmen, n.d.)
- **Authentication and Authorization:** Implement multi-factor authentication (MFA) and role-based access control (RBAC) to secure access to user accounts and administrative functions.
- **Compliance:** Ensure the app complies with GDPR, HIPAA, and other relevant data protection regulations to protect user privacy.
- **Vulnerability Management:** Regular vulnerability assessments and penetration testing must be conducted, with a plan for prompt patching of any discovered vulnerabilities.

4. Usability Requirements

- **User Interface:** The app should have an intuitive and user-friendly interface, particularly for mood logging, journaling, and accessing guided meditations.
- **Accessibility:** Ensure the app is accessible to users with disabilities, complying with WCAG 2.1 guidelines stipulated by the Singapore Government developer Portal to provide an inclusive user experience. (*Web Content Accessibility Guidelines (WCAG) 2.1*, 2023)
- **Documentation:** Provide comprehensive user documentation, including FAQs, help guides, and tutorial videos to assist users in navigating the app.
- **Localisation:** Support for multiple languages, initially only English, followed by Chinese, Malay and Tamil for local Singaporean users.

5. Maintainability Requirements

- **Code Quality:** Adhere to coding standards and best practices, with code reviews and automated testing integrated into the development process to ensure maintainability.
- **Modularity:** The system should be modular to allow for easy updates and integration of new features such as additional mental health resources or enhanced AI capabilities.
- **Logging and Monitoring:** Implement comprehensive logging and monitoring to detect and address issues proactively, particularly in critical features like mood tracking and journaling.
- **Documentation:** Maintain detailed technical documentation for the system architecture, APIs, and codebase to assist future development and maintenance efforts.

6. Interoperability Requirements

- **API Integration:** The app should support integration with third-party APIs (e.g., for guided meditations, therapist sessions, and community forums) to enhance functionality.
- **Data Import/Export:** Allow for easy import and export of user data in standard formats (e.g., JSON, CSV) to facilitate data portability.
- **Platform Compatibility:** Ensure compatibility across different platforms (iOS, Android, Web) and devices (mobile, tablet, desktop) to provide a consistent user experience.

7. Efficiency Requirements

- **Resource Utilisation:** The app should efficiently use CPU, memory, and network resources, minimising battery consumption on mobile devices to enhance usability.
- **Bandwidth Usage:** Optimise data transmission to minimise bandwidth usage, especially for users with limited connectivity, ensuring smooth access to all app features.

8. Portability Requirements

- **Deployment:** The app should be easily deployable on different cloud platforms (e.g., AWS, Azure, GCP) to provide flexibility in infrastructure choices.
- **Cross-Platform Support:** Develop using cross-platform frameworks (e.g., React Native) to ensure consistency across different operating systems and devices.

9. Compliance and Legal Requirements

- **Data Privacy:** Ensure all user data is collected, stored, and processed by relevant privacy laws and regulations, providing transparency and trust.
- **Intellectual Property:** Respect and protect intellectual property rights, ensuring all content and software components are properly licensed.

10. Environmental Requirements

- **Operating Environment:** The app should function reliably under various network conditions, including low bandwidth and intermittent connectivity, ensuring accessibility in diverse environments.
- **Compatibility:** Ensure the app runs smoothly on devices with different hardware specifications, including older models, to reach a broader audience.

Form - Vision & Design

Name

SAGE

The name "SAGE" was selected to evoke wisdom and guidance, aligning with the app's role as a trusted companion in navigating mental health challenges. Its design emphasises inclusivity, with a user-friendly interface tailored to accommodate diverse needs and backgrounds, ensuring everyone can benefit from its resources and support.

Acronym

The acronym "SAGE" encapsulates the app's holistic approach to mental health support: Supportive, Accessible, Guiding, and Empowering. These principles were carefully chosen to reflect our commitment to providing empathetic support, ensuring accessibility for all users, guiding individuals towards mental well-being through personalised plans, and empowering users to take control of their mental health journey.[6]

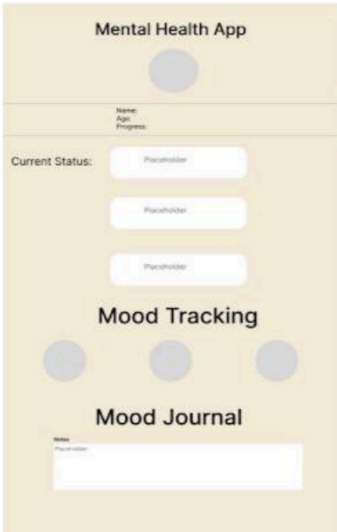
Logo

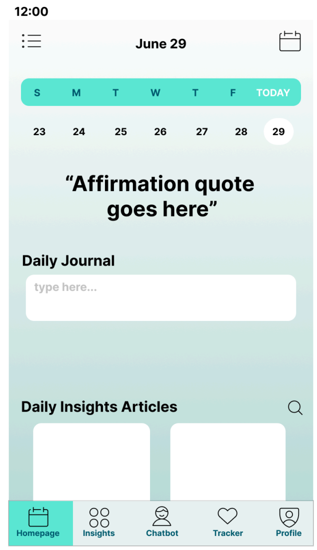
Our logo embodies two symbols: an owl representing wisdom and guidance in mental health, and an angel symbolising compassion and supportive care. Together, they reflect our app's mission to provide insightful guidance and empathetic support, guiding users on their journey to mental well-being and offering comfort during difficult times.

Colour Scheme

Following user feedback, we reevaluated our initial colour scheme of beige and light purple with black font due to concerns about its perceived lack of warmth and relaxation. Responding to this feedback, we refined our design approach by creating a second wireframe using soothing colours like sea green and blue. These selections were deliberate to cultivate a tranquil environment, enhancing the app's appeal as a calming and supportive resource for mental well-being. These colours will be used for the prototype.

[6] Please refer to appendix D for a detailed table of the acronym


Mental Health App
Current Status:
Mood Tracking
Mood Journal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Literature

The mental health landscape has witnessed a significant shift with the advent of digital interventions, particularly mental health companion apps. These applications aim to provide support for individuals experiencing depression and anxiety by offering a range of functionalities, including mood tracking, therapeutic exercises, and access to informational resources and community support. The following literature review highlights key research in the field, underscoring the relevance of these studies to our project, which aims to develop a comprehensive and user-centred mental health companion app tailored for young adults in Singapore.

Prevalence of Mental Health Issues in Singapore

In Singapore, the **2016 Singapore Mental Health Study** revealed that one in seven individuals had experienced a mental disorder in their lifetime, with major depressive disorder and generalised anxiety disorder being the most prevalent. More recently, the **National Population Health Survey of 2022** highlighted a concerning rise in poor mental health among Singapore residents aged 18 to 74, increasing from 12.5% in 2017 to 17.0% in 2022. This increase is particularly pronounced among youths aged 18 to 29, with prevalence rates escalating from 16.5% in 2017 to 25.3% in 2022 **(Epidemiology & Disease Control Division, Policy, Research & Surveillance Group, Ministry of Health, and Health Promotion Board, Singapore, 2022).** This significant increase underscores the pressing need for effective mental health interventions, especially for young adults in Singapore. Based on our research, we have identified our primary target market for the mental health companion app to be individuals aged 18 to 29 who are dealing with depression or anxiety-related issues.

Privacy and Security Concerns

One of the major concerns with mental health apps is the privacy and security of user data. Studies highlight the need for robust encryption and secure communication channels to protect sensitive information. Ensuring compliance with the **Personal Data Protection (Amendment) Act 2020** is crucial to the integrity of our app. However, there are still several challenges we will face while developing the mental health companion app.

Vulnerability to Data Privacy Breaches

Our mental health companion app logs highly sensitive and private information from users, including their moods and thoughts. A recent study on the privacy of mental health apps underscores that these applications are "*especially vulnerable to data privacy breaches as a result of security attacks*," with a "*high prevalence of information disclosure threats, mainly originating from insecure programming*" (Iwaya et al., 2022). Therefore, ensuring the security of the data we collect is imperative. This involves implementing robust encryption methods, adhering to the latest security protocols, and conducting regular security audits to identify and mitigate potential vulnerabilities. By prioritising data protection, we aim to maintain user trust and comply with stringent privacy regulations.

Ethical Consideration

Moreover, ethical considerations must be constantly focused due to the nature of our application and its users. Patients "*need to be fully aware and consent to how their data are collected, managed, and used*" (Smith et al., 2024). This is particularly crucial for our AI features, which rely significantly on user-provided data. We must ensure that users are fully informed and have agreed to the data usage policies. Additionally, when designing the app's features, it is essential to ensure that "*digital interventions are based on high-quality evidence from diverse populations, are predictively valid and support accurate diagnoses and prognoses, and do not themselves cause harm*" (Smith et al., 2024). This consideration, quoted from a panel of experts on digital mental health, emphasises the need for our app to be beneficial without triggering or exacerbating existing issues. Ensuring the app's interventions are safe and effective is a key priority.

Proven Effectiveness of Similar Tools in Current Market

When examining existing mental health apps, a summary article indicates that most apps prioritise informing, recording, and collecting information, providing users with valuable insights into their mental health (Myers et al., 2021). Specifically, features that have proven effective include those seen in Woebot, a therapy chatbot utilising Cognitive Behavioral Therapy (CBT). A study on Woebot demonstrated favourable results, with users experiencing reduced depression compared to non-users. The study highlighted key features such as daily check-ins for accountability, the bot's empathy and personality, and the insights provided by the bot (Fitzpatrick et al., 2017). In the context of our target group, Singaporeans, a previous study involving university students identified the most desired features in a mental health app. These features included personalisation, low or no cost, an aesthetically pleasing and user-friendly interface, and backing by research and endorsements from trusted parties (Sanabria et al., 2023).

Therefore, for our app, we will incorporate successful elements from other applications and consider feedback from previous studies to create a more effective tool for our users.

Market Research

App Market Overview

The mental health app market is experiencing significant growth, reflecting broader trends in mental health awareness and technology adoption. In 2023, the market was valued at $6.2 billion and is projected to grow at a compound annual growth rate (CAGR) of 15.2% from 2023 to 2030. (*Mental Health Apps Market Size And Share Report, 2030*, n.d.)

Several key drivers underpin this growth – the increasing prevalence of depression, anxiety, and other mental health disorders has been exacerbated by the COVID-19 pandemic, which induced feelings of isolation and uncertainty across society. The widespread use of smartphones has made these apps more accessible to a larger population, facilitating their adoption. Growing recognition and awareness of mental health issues, fueled by social discussions and advocacy, have also contributed to the demand for these digital solutions. Moreover, the heightened need for remote mental health support due to COVID-19 quarantine restrictions has further accelerated the market's expansion.

This convergence of factors highlights the critical role of mental health apps in addressing contemporary mental health challenges.

Key Factors Driving Market Growth

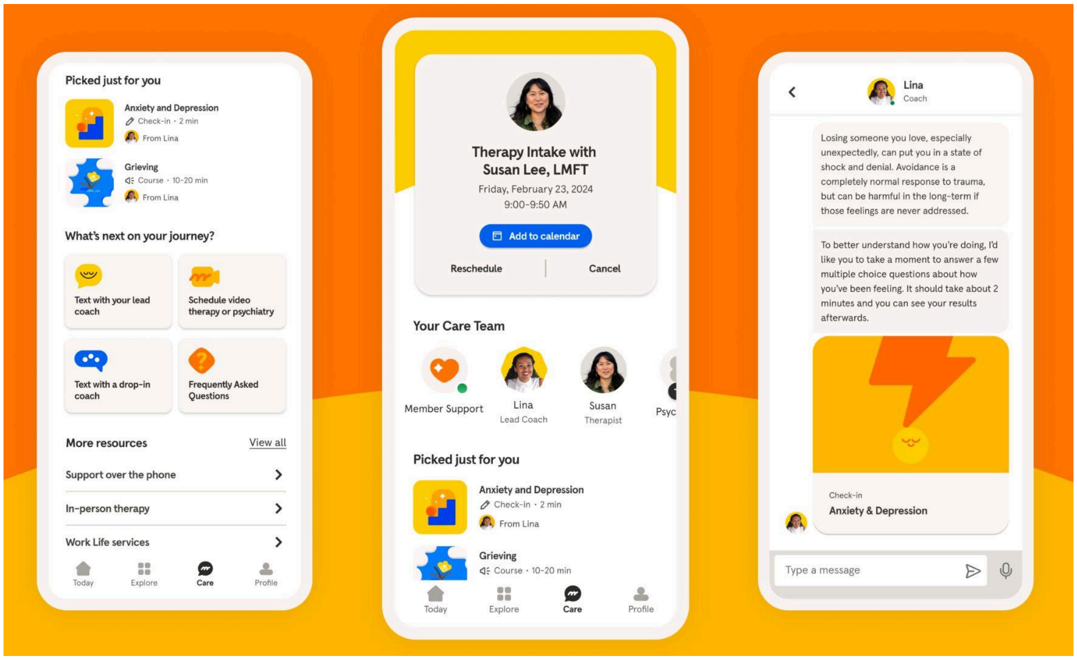
Our market research indicates several key factors driving the growth of mental health companion apps. Firstly, there is a greater public understanding of the importance of mental health, largely due to social efforts aimed at de-stigmatizing mental health misconceptions. This has led to increased openness and willingness to seek help. Additionally, mobile health technologies, particularly telehealth, are becoming more common, providing a convenient and accessible option for many users. The high costs of traditional mental health services further push the demand for affordable mobile solutions, making mental health apps an attractive alternative.

Furthermore, the COVID-19 pandemic has significantly accelerated the need for remote support. Social distancing measures and the increased stress and anxiety during the pandemic period have underscored the importance of accessible mental health resources, driving more people towards these digital solutions.

According to a report by Grand View Research, the mental health apps market is expected to grow significantly due to these factors (*Mental Health Apps Market Size And Share Report, 2030*, n.d.).

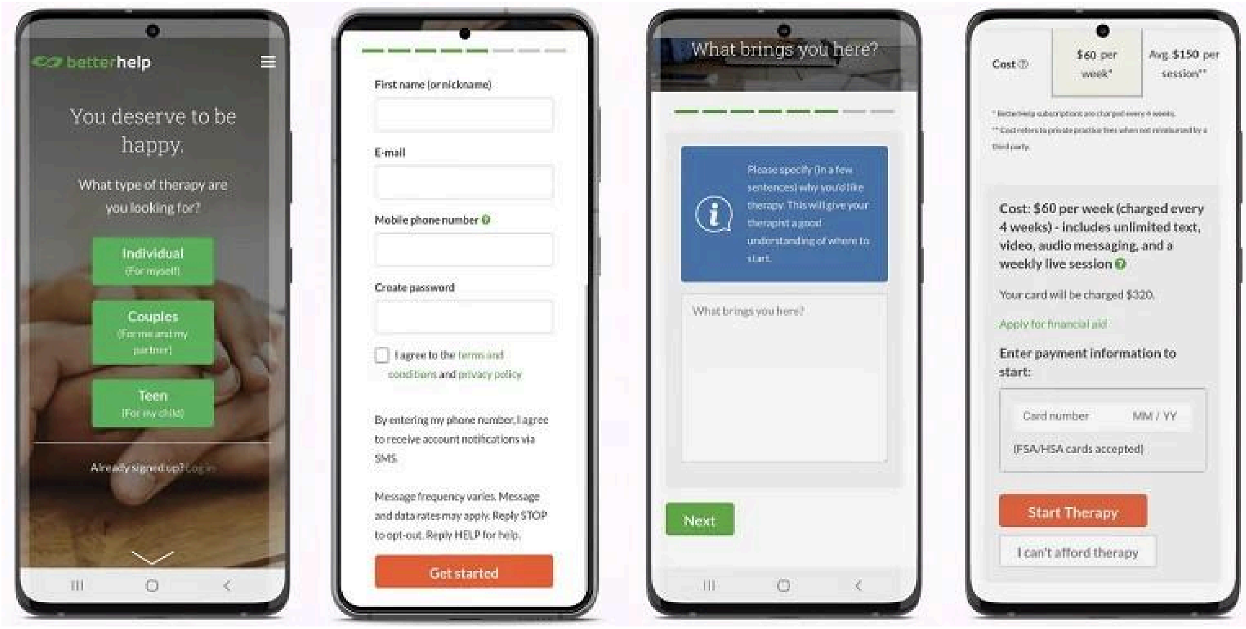
Key Competitors

Headspace



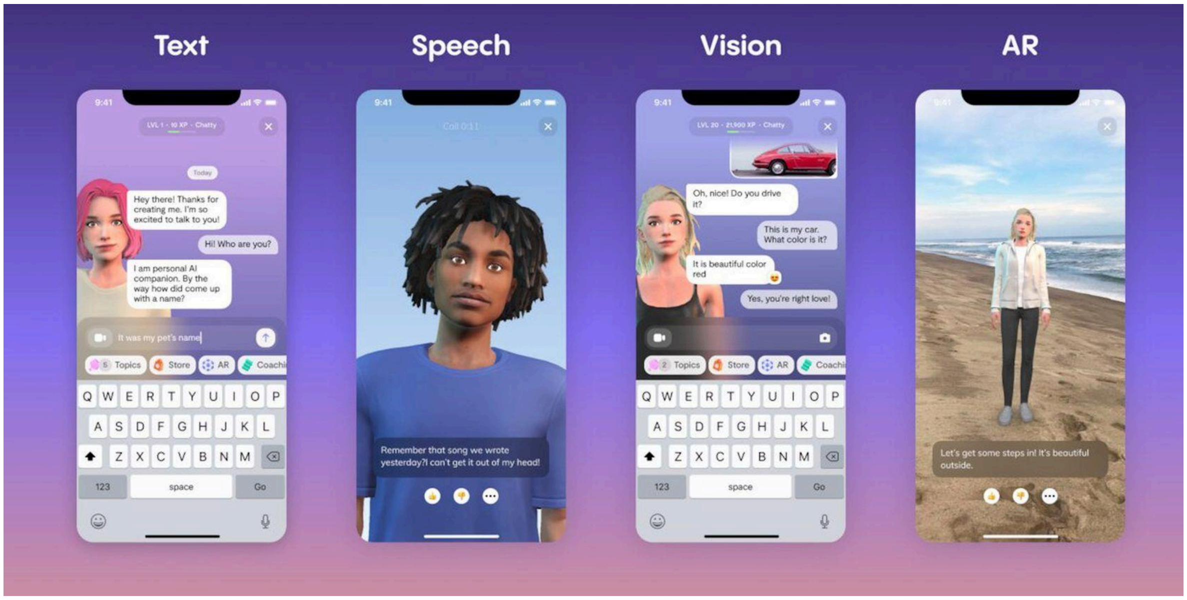
- **Description:** Mindfulness and meditation app with guided meditations, sleep aids, and wellness programs.
- **Market Share & User Base:** 15% (Sensor Tower). Over 65 million downloads globally.
- **Revenue Model:** Subscription-based, corporate programs, and branded partnerships while offering free content to users
- **Strengths:** Strong brand, high-quality content, user-friendly interface, wide range of features.
- **Weaknesses:** Limited mood tracking and personalised support, premium subscription costs

BetterHelp



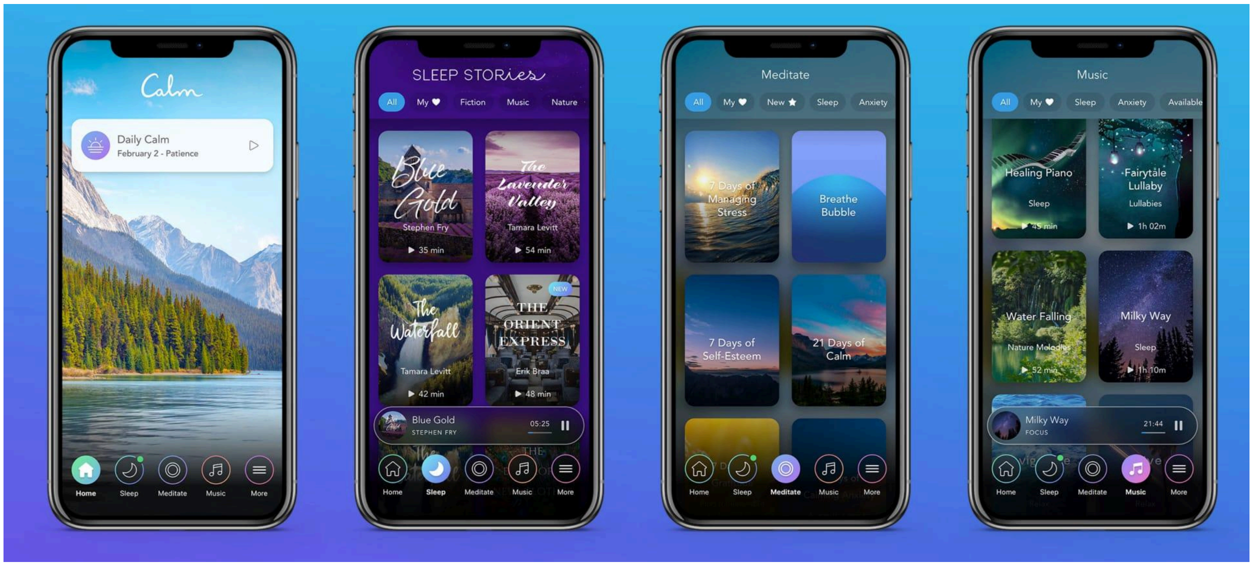
- **Description:** Online therapy through licensed counsellors via text, phone, and video sessions.
- **Market Share & User Base:** Leading online therapy platform with over 1 million users.
- **Revenue Model:** Subscription-based. Estimated >$1 billion annual revenue (*BetterHelp Analysis & Market Share Overview*, n.d.)
- **Strengths:** Access to professional therapists, flexible communication, comprehensive services.
- **Weaknesses:** Higher costs compared to self-help apps, privacy concerns, not suitable for severe conditions.

Replika



- **Description:** AI chatbot for emotional support and companionship.
- **Market Share & User Base:** AI-driven with over 10 million downloads.
- **Revenue Model:** Subscription-based and microtransactions with gamification incentives
- **Strengths:** Personalised interactions, available 24/7.
- **Weaknesses:** Limited therapeutic capabilities, data privacy and AI accuracy concerns.

Calm



- **Description:** Focuses on meditation, sleep, and relaxation with guided sessions, sleep stories, and music.
- **Market Share & User Base:** Leading meditation and mental wellness app as of January 2023 (Ceci, 2023); 4 million paying subscribers in November 2022 (Southwick, 2022)
- **Revenue Model:** Subscription-based, corporate and team programs, and branded partnerships while offering free content to users.
- **Strengths:** High-quality content, appealing design, strong focus on sleep and relaxation.
- **Weaknesses:** Limited interactive features, and high premium subscription costs.

The mental health app that we have planned offers a comprehensive set of features that caters to a wide range of user needs. So, we have envisioned an app that integrates the best features identified in our market research. Like “Headspace” and “Calm”, our app also offers useful guided meditation and articles but we go further by including mood tracking and goal settings, which we think the other apps lack. Other than features from “BetterHelp” which provides online therapy and “Replika” that provides AI chatbot for emotional support and companionship, we also include Journaling with AI prompts that promotes self reflection and mental growth. Lastly, we included a community forum in our features to promote user interaction and peer support, which we see lacking in the other competitors from our market research.

User Needs and Preferences

User Demographics

- **Age Group:** Primarily young adults (18-55 years), usage across all age groups.
- **Gender:** Fairly balanced, higher percentage of female users in some studies.
- **Geographical Distribution:** High usage in urban areas with good internet connectivity, higher adoption in developed countries.

Key User Needs

- **Privacy and Security:** High concern for data privacy and security.
- **Ease of Use:** Preference for intuitive interfaces and easy navigation.
- **Personalisation:** Tailored content and features.
- **Accessibility**: Affordable or free options, offline features availability.
- **Comprehensive Support:** Integration of mood tracking, daily affirmations, professional support options, and crisis management.

Market Trends

AI and Machine Learning

- **Trend:** Use of AI for personalised recommendations, mood tracking, and predictive analytics.
- **Example:** Woebot, an AI chatbot, effective in managing anxiety and depression.

Integration with Wearable Devices

- **Trend:** Wearables like smartwatches and fitness trackers monitor physiological indicators related to mental health.
- **Example:** Apple Health integrates mental and physical health data.

Focus on Data Privacy

- **Trend:** Enhanced privacy and security due to user concerns and regulatory requirements (e.g., GDPR, HIPAA).
- **Example:** End-to-end encryption and transparent data policies.

Holistic Health Approaches

- **Trend:** Combining mental, physical, and emotional well-being (e.g., fitness tracking, nutrition advice, sleep management).
- **Example:** MyFitnessPal integrates physical health tracking with mental health features.

Teletherapy and Online Counseling

- **Trend:** Growth in teletherapy and online counselling services.
- **Example:** BetterHelp and Talkspace expand services to meet demand for online therapy.

Potential Opportunities

Gap Analysis

Prioritising privacy and data security through transparent data handling and robust security measures is crucial. Comprehensive features, such as all-in-one solutions that include affirmations, mood tracking, predictive analytics, and professional support, can enhance user experience. Affordability is another key opportunity, with affordable pricing models or freemium options appealing to a broader audience. Additionally, incorporating crisis management features that provide immediate support during severe mental health episodes can significantly improve the value and effectiveness of these apps.

Differentiation Strategies

Differentiation strategies in the mental health companion apps market can drive competitive advantage. Emphasising user-centric design ensures an intuitive and accessible user experience. Integrating advanced AI capabilities allows for accurate mood tracking and predictions through AI and machine learning. A holistic health approach, which combines physical health tracking, nutrition advice, and sleep management, offers a more comprehensive wellness solution. Additionally, forming partnerships with health professionals can provide credible and reliable content and support services, enhancing the app's trustworthiness and effectiveness.

Approach & Motivations

User-Centred Design (UCD)

The primary motivation for adopting a User-Centred Design (UCD) approach is to ensure that the mental health companion app effectively addresses its users' specific needs and preferences. By placing users at the heart of the design process, we can create a product that is intuitive, accessible, and valuable to those dealing with depression and anxiety.
UCD also helps develop a user-friendly interface that accommodates users with varying levels of technical proficiency and different accessibility needs, which is crucial for a mental health app.The iterative nature of UCD allows for ongoing testing and refinement based on user feedback, leading to continuous enhancements in the app's design and functionality.

Development

Our development roadmap for the mental health companion app focuses on key deliverables and tasks to create a robust platform. We start with the **SCRUM framework** set-up, prioritising initial user interviews and user story gathering to inform feature development. Key features include mood tracking, therapist communication, an AI chatbot, and AI-driven journaling designed to personalise and enhance user experiences. UI/UX efforts concentrate on intuitive design for mood-logging and trend visualisation. Guided meditations tailored for depression and anxiety complement these features.

MoSCoW Method

In planning the development and release of our mental health companion app, we employed the MoSCoW method to prioritise features and ensure that the most critical functionalities are delivered first. The MoSCoW method, which stands for Must have, Should have, Could have, and Won't have, allowed us to systematically categorise our user stories and requirements based on their importance and impact on the user experience.

Key features like mood tracking, AI-driven journaling, and the AI chatbot were classified as "Must have" and scheduled for development in the initial sprints. Following this, features deemed "Should have," such as goal setting and progress tracking, guided meditations, and secure messaging with therapists, were planned for subsequent sprints..

Optional enhancements, labelled as "Could have," like theme customisation, are planned for later sprints. Features categorised as "Won't have" for the initial release will be considered for future updates.

By methodically prioritising through the MoSCoW framework, our app development remains user-centric, addressing the most crucial needs first while allowing for flexibility and continuous improvement.

SCRUM (Agile-based framework)

We chose Scrum for our app development to consistently deliver user-centred updates through structured, milestone-aligned sprints. Scrum's iterative nature and regular stakeholder feedback loops enhance functionality, usability, and responsiveness to user needs and market trends.

This approach allows us to break down our development into manageable sprints, each focused on delivering specific features or improvements, starting with the Mood Tracker feature. This aligns perfectly with our goal of achieving a beta release by November 18th, 2024.

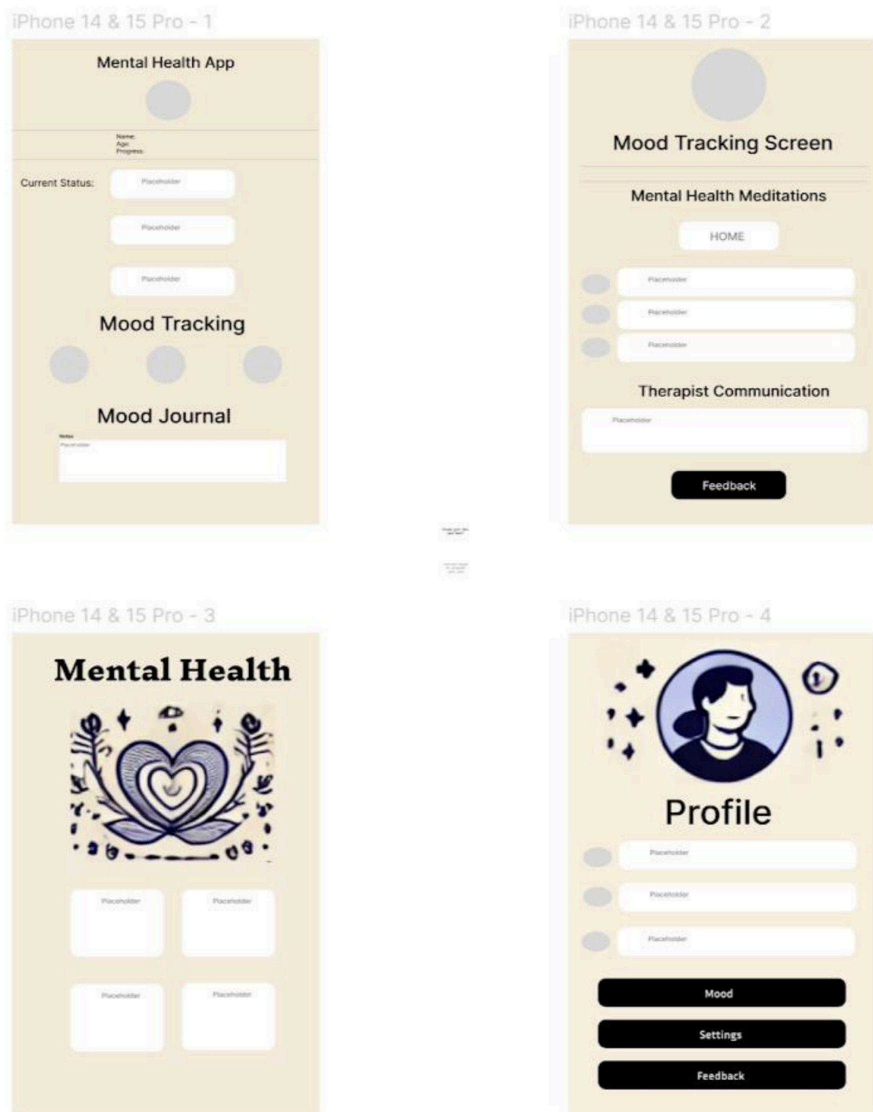
Tools

In our project, we use tools designed to enhance collaboration, streamline development, and effectively gather user feedback. These tools cover various aspects such as communication, prototyping, user testing, documentation, and version control. Here's the list of tools used:

- **Meetings:** Telegram, Discord
- **Prototyping and Design:** Figma and Canva
- **Survey and User Testing:** Google Forms
- **Notes and Proposal:** Google Docs
- **Data Sharing & Documentation**: Google Drive and Github
- **Code tracking/review & Version Control:** Github

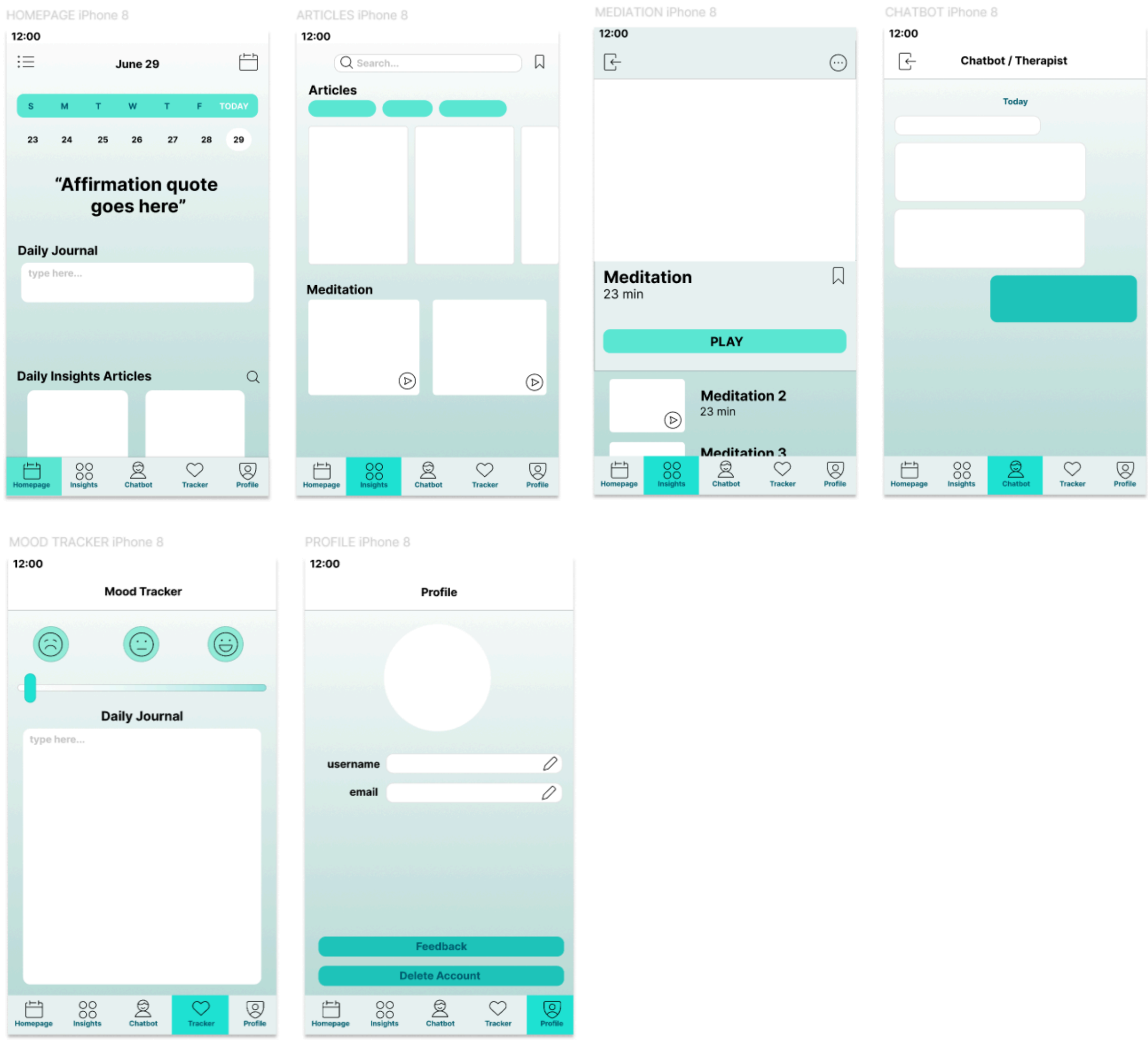
Prototyping & Iterative Development

Initial Prototype (Wireframe) (Stage 1)



The initial prototype of our mental health app shows some of the pages that we intend to have in our app. The wireframe focused more on showing the features that we wanted to have in the app: mood tracking, journaling, guided meditations etc.keeping the different features of the app easily identifiable at first glance. However, user feedback said colours of the pages were not appealing, and the wireframe was not showing a clear enough detail of the app. The pages also felt disconnected.

Low-fidelity Prototype (Stage 2)



Hence, we have improved the wireframe by adding some features to make the app look more intuitive and easier to navigate.

One of the features is the bottom navigate bar, it aligns with the thumbs rule design, the principle that most users navigate apps using their thumb. This makes navigating the app easier for the user. It also includes a more personalised main screen and a preliminary layout for the guided meditation section. The wireframe shows a better visualisation of the app features.

Assumption testing

Assumption testing in our mental health companion app development follows a **user-centred design (UCD) approach**, ensuring that our initial hypotheses align closely with **user needs and expectations.** By systematically validating these assumptions through iterative testing and user feedback, we aim to mitigate risks early and refine the app's usability and functionality. This approach informs strategic decisions and fosters a deeper understanding of user behaviours, guiding us towards creating an intuitive and impactful tool for mental well-being.

As no hard coding has been done at this point of the project, we can only test our assumptions based on the app concept and initial wireframes designed using low-fidelity methods. They are as follows:

Surveys and Interviews

To gain insight into the stakeholders, we considered the largest stakeholders of the project and designed a set of interviews for each stakeholder and interviewed them. Our aim was to get insight from two potential end users, a mental health professional and a data protection officer, as we understand the highly sensitive nature of data that would be collected and value its privacy and security. The questions in Appendix A were asked during the interview and grouped by stakeholder (Tables A1, A2 and A3)

We also designed a survey and disseminated it to our connections for a wider audience reach. The questions and their related options contained in Appendix C were asked in the survey.

The following were the assumptions we made about the features of the app, and also the validation that users have given us through the surveys and interviews:

Assumptions	Validation
Users need a simple and intuitive way to track their moods daily	**Guided meditations** and **mood tracking** were consistently mentioned as valuable features.
Users prefer open-ended journaling prompts that are relevant to their mental health	**Common dislikes** included paywalls, repetitive content, generic responses from chatbots, and a lack of in-depth solutions.
Secure communication with therapists is crucial for user trust	**Therapist communication** was also considered important by multiple subjects.

	All subjects expressed **high concern about the privacy and security of their data** when using mental health apps.
The app's interface should be easy to navigate, even for users with low technical proficiency	A preference for **clear, simple**, and **minimalistic** interfaces, with easy navigation and minimal complexity highlighted as important features.
Users will engage more with personalised content and recommendations	While **personalised** content and recommendations were generally seen as useful, they should be **optional** and **non-intrusive**.While **personalised** content and recommendations were generally seen as useful, they should be **optional** and **non-intrusive**.
The app should be accessible to users with varying levels of visual and cognitive abilities	While specific accessibility needs were not highlighted by most interviewees, **ensuring readability** and **easy-to-use features** was mentioned.

Conclusion:

The responses collected from the surveys and interviews validated most of the assumptions made about the app features and user interface.

Wireframe Testing[7]

As wireframes are low-fidelity likenesses of user interfaces, it was the fastest and easiest way for us to get feedback on some assumptions we hold regarding the app's information structure and functionality. Wireframes also allow us to design our app in many quick iterations, which enable swift identification of flaws in designs that are unlikely to succeed, allowing for timely adjustments and improvements.

We did a 5-second test to find how users remember our designs after five seconds by showing them the first iteration of the wireframes, based on the assumption that the design and layout were simplistic yet appealing, with clear navigation layout. However, the assumptions were wrong. Users feedback that the colour scheme was not inviting, too drab and not soothing for a mental health companion app. Buttons and features were not aligned and labels were lacking in some parts of the layout. Gathering these responses, we made some changes to the

[7] Please refer to the Prototyping & Iterative Development section for diagrams of wireframes.

wireframes and are ready for the second iteration to be tested further with usability testing instead.

Analysis & Outcomes

In conclusion, our journey towards developing the mental health companion app has been challenging and enlightening. On the one hand, we encountered several failures. We lacked a concrete plan structure before embarking on the proposal, which highlighted the need for better planning of task delegation and time management. Our initial approach of working on each section individually, without considering their interdependence, compromised the proposal's flow and caused segmentation. Additionally, our understanding of the agile framework remains unclear, and the lack of professional and expert insight due to unavailable resources further hampered our progress.

Despite these setbacks, we achieved significant successes. We have a clear understanding of what we want to create, the features it will include, and how we aim to impact the community we have chosen to serve. Our team is empathetic and understanding of the user space, with at least two out of five teammates personally benefiting from the app. This personal connection fuels our commitment to developing a user-centric product. We also have a clear UI design, which we plan to refine based on user feedback.

Currently, our project is still in its nascent stages. Prototypes have yet to be issued to the first batch of user testers, preventing us from receiving critical feedback on the current prototype. Additionally, we have yet to attempt any hard coding. Moving forward, it is imperative that we address these issues, improve our planning and understanding of the agile framework, and seek expert insights to ensure the success of our project.

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Appendix

Appendix A

Table A1

Patients (Users)
Have you used any mental health apps before? If so, what did you like or dislike about them?
Which features would you find most valuable in a mental health companion app? (e.g., mood tracking, journaling, guided meditations, therapist communication)
How important is it for you to have personalised content and recommendations?
What are some characteristics of an app that you find user-friendly?
Do you have any specific accessibility needs that should be considered in the app design?
How concerned are you about the privacy and security of your data when using a mental health app?
What measures would make you feel more secure about your personal information being protected?
What would motivate you to use the app regularly?
Are there any particular types of notifications or reminders that you find helpful?

How important is it for the app to provide educational content and resources on mental health?
Would you be interested in features that allow for social support or community interaction within the app?
How would you prefer to provide feedback on the app's features and usability?
Are there any additional features or services you would like to see in a mental health companion app?

Table A2

Mental health professional
Can you describe your role and experience in the mental health field?
What common challenges do your clients face that could be supported by a mental health companion app?
Which app features do you think would be most beneficial for your clients? (e.g., mood tracking, journaling, guided meditations, therapist communication)
How important is it to have secure communication channels between therapists and clients within the app?
How can a mental health app complement the therapy sessions you provide?

What types of data or insights from the app would be useful for you in supporting your clients?
What ethical considerations should be taken into account when developing and using a mental health companion app?
How can we ensure that the app adheres to professional standards and guidelines?
What strategies do you recommend for keeping users engaged with the app and adhering to their mental health routines?
How can the app support clients between therapy sessions?
How would you like to be involved in the ongoing development and improvement of the app?
Are there any specific features or tools you believe are missing in existing mental health apps?

Table A3

Data protection officer
Can you describe your role and experience in data privacy and security?
What are the key data privacy regulations and standards that the app needs to comply with (e.g., GDPR, HIPAA)?
How can we ensure that the app meets these regulatory requirements?

What types of personal data should the app collect, and what should be avoided?
How should the app securely store and manage user data to prevent unauthorised access?
What best practices should be followed to obtain user consent for data collection and usage?
How can we ensure that users are fully informed about how their data will be used and protected?
What specific security measures should be implemented to protect user data (e.g., encryption, access controls)?
How can we regularly audit and update these security measures to address evolving threats?
What should be included in an incident response plan in case of a data breach?
How can we communicate effectively with users in the event of a data security incident?
How can we provide users with control over their personal data (e.g., data access, correction, deletion)?
What processes should be in place to handle user requests related to their data rights?
How would you recommend we gather feedback on our data privacy practices from users and stakeholders?

Are there any emerging data privacy trends or technologies that we should consider incorporating into the app?

Appendix B

User Stories

Feature	User Stories
Mood Tracking	As a user, I want to log my mood daily so that I can track changes in my emotional state over time.
	As a user, I want to view visualisations of my mood trends so that I can identify patterns and triggers.
Goal Setting/ Progress Tracking	As a user, I want to set personal mental health goals so that I can work towards improving my well-being.
	As a user, I want to track my progress towards these goals to stay motivated and see my improvements.
Personalised Content	As a user, I want to receive personalised recommendations for articles and exercises based on my mood logs and preferences so that I can find relevant resources easily.
	As a user, I want the app to provide daily motivational quotes and reminders to help keep me encouraged.
Guided Meditations	As a user, I want access to guided meditations tailored to my specific needs, such as anxiety or depression, so that I can practise mindfulness and relaxation techniques.
	As a user, I want to set reminders for meditation sessions to maintain a consistent practice.
Journalling	As a user, I want to use an AI-driven journaling feature that suggests prompts to help me reflect on my thoughts and emotions.
	As a user, I want to review my past journal entries to gain insights into my emotional journey.
Community Support	As a user, I want to connect with a supportive community within the app to share my experiences and seek advice.
	As a user, I want to participate in discussion forums and group activities to feel less isolated.
Crisis Management	As a user, I want to have quick access to emergency contacts and crisis resources within the app in case I need immediate help.
	As a user, I want the app to provide calming techniques and immediate support during moments of high anxiety or panic.
Professional	As a user, I want to find information about local mental health professionals and support services so that I can seek professional help if needed.

Feature	User Stories
Resources	As a user, I want to access educational content about mental health conditions to better understand my own experiences and treatment options.
Customisation	As a user, I want to customise the app's interface and features according to my preferences to enhance my user experience.
	As a user, I want the option to choose different themes and layouts to make the app visually appealing to me.
Reminders and Notifications	As a user, I want to set reminders for taking medications, attending therapy sessions, or performing daily exercises to ensure I stay on track with my mental health plan.
	As a user, I want to receive notifications about new content and community events within the app.

Appendix C

- Age:
    - 18-24
    - 25-34
    - 35-44
    - 45-55
    - 55+
- Gender:
    - Male
    - Female
    - Non-binary
    - Prefer not to say
- Location:
    - Singapore
    - Other (please specify)
- Occupation:
    - Student
    - Employed

    - Self-employed
    - Unemployed
    - Retired
- How important is it for you to track your daily mood in a mental health app?
- How likely are you to use a journaling feature that provides AI-generated prompts?
- How useful would you find a library of guided meditation sessions in the app?
- How important is the ability to communicate securely with your therapist through the app?
- Would you participate in social support groups within the app to share experiences and support others?
- How beneficial would it be to have a feature that allows you to set and track personal mental health goals?
- Would you like the app to integrate with wearable devices to monitor your physical activity and sleep patterns?
- How important is it for the app to provide articles, tips, and videos on managing mental health?
- How important is an easy-to-navigate interface for you in a mental health app?
- Do you have any specific accessibility needs (e.g., visual impairments) that should be considered in the app design? If yes, please specify.
- How important is the speed and responsiveness of the app?
- How important is the ability to customize the app, its features and interface to suit your preferences?
- How important is having access to customer support or help resources within the app?
- Are there any other features you would like to see in a mental health companion app?
- Do you have any concerns or additional suggestions for the app, it’s design and functionality?

Appendix D

S - Supportive	**A -** Accessible	**G -** Guiding	**E -** Empowering
• Offers empathetic and non-judgmental support. • Provides resources for mental health issues. • Encourages users through difficult times with positive reinforcement.	• Available 24/7 for *immediate assistance.* • User-friendly interface suitable for all age groups. • Inclusive design accommodating diverse needs and backgrounds.	• Offers personalised mental health plans and suggestions. • Guides users through meditation, mindfulness, and coping strategies. • Provides a structured path to achieve mental well-being goals.	• Empower users to take control of their mental health. • Encourages self-awareness and self-care practices. • Provides tools and resources to help users build resilience and confidence.

Appendix E (Meeting Notes & Initial Planning)

29 June 2024

Lecturer's comment

1. Try to keep things in paragraph format instead of point form
2. Do not use pictures and tables as main content. Can have those but need to elaborate with paragraphs. Pictures and tables are better to be used as reference items.
3. Edit the sprint part based on release-based (agile) 8:59
4. Wireframe needs to be more elaborate (go into key features, have e.g. on what the bot does), add branching of the app(?)

Actionable items (what, who, when)

1. Change point form to paragraphs, edit according to feedback (everyone, deadline: 2nd july)
2. Elaborate/change the pictures and tables part. (Main part: Planning area (the table), stakeholders (the picture)
3. Elaborate/add detail to wireframe (xiang lin, 3rd july)

19 June 2024

Attendees: All

Notes

- Journaling is a very useful feature, according to an Occupational Therapist (source: Xiang Lin's sister)
- Feature suggestion: Auto-integration / shortcuts for in-app features like journalling (Customizable?)
- PS interviews are done for now (5/30), continue after Mid-Term submission (8th July)
- Next meeting 26th June
- PESTLE & SWOT analysis excluded from Approach and Motivations (already included in Market research)

Action items

- [x] ~~Put geographic scope in Aims & Objectives and focus on technical scope in Scope~~
- [x] ~~Do a Gantt chart (do a general one maybe for 9 weeks, 30th June - 2nd September)~~
- [x] ~~*SuiRong, HuiToon:* Wireframe and prototypes~~
- [x] ~~*Devon, Mandy:* Update to requirements based on interview and survey analysis~~
- [x] ~~*Mandy:* Conduct the last interview for proposal, extract and analyse interview responses~~

- [x] ~~26th JUNE: Everything EXCEPT 'Prototyping and Iterative Development', 'Assumption testing', 'Analysis and Outcomes' should be done~~

10 & 15 June 2024
Attendees: All
Notes

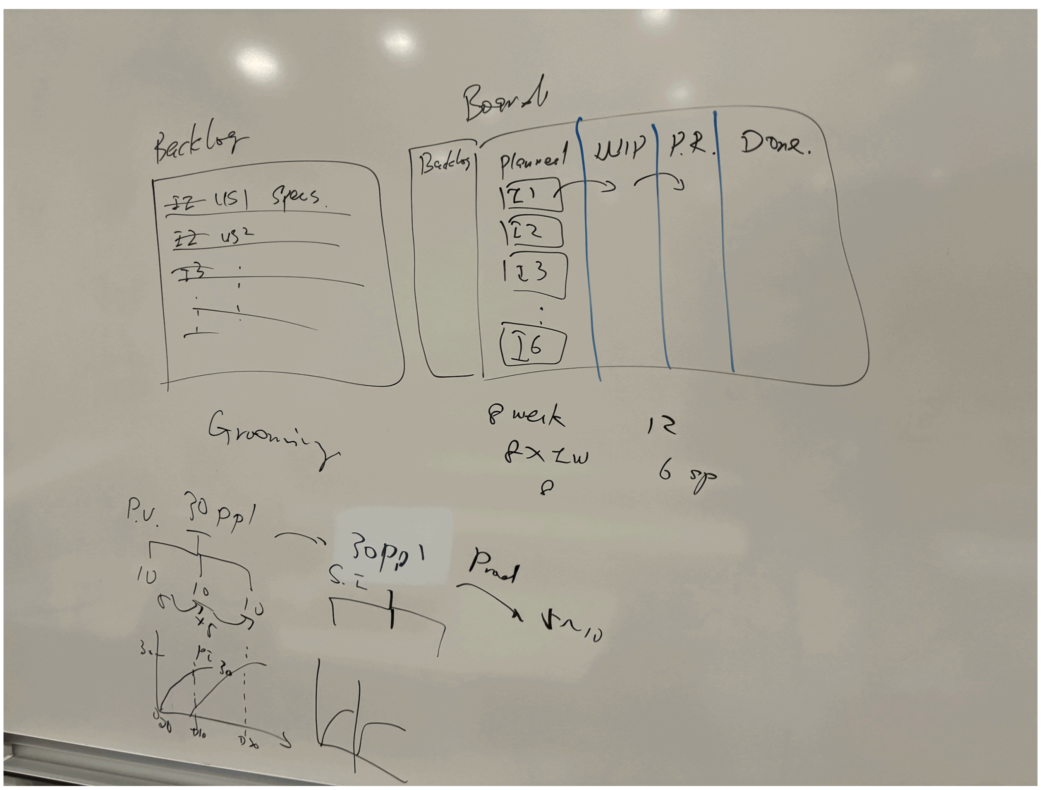
- Timeline context: Can do until any stage i.e., deployment but need to put marker on where we're stopping for this project, and include that in the scope section of proposal.
- Using SCRUM: https://www.scrumalliance.org/about-scrum?utm_source=badgecert.com&utm_medium=social&utm_campaign=bdg-sharedigitalbadge&utm_content=scrum-artifacts
- Retirement age is 63 currently, will be raised to 64 in 2026, re-employment age to rise t0 69 (Source: https://www.straitstimes.com/singapore/politics/s-pore-retirement-age-to-go-up-to-64-in-2026-re-employment-age-to-rise-to-69)
- According to this study, the younger population are more at risk for both depression and anxiety, the study was broken up into 3 age groups, i.e., 21-39, 40-64 and >=65, with the prior having the largest prevalence in our selected 2 illnesses.(https://annals.edu.sg/prevalence-and-risk-factors-of-depression-and-anxiety-in-primary-care/#:~:text=Younger%20age%20groups%20exhibited%20a,those%20aged%20%E2%89%A565%20years.)
- Interview sample size for Problem Statement ~30, start with 5-10 for proposal first
- Interview sample size for Solution Statement ~30 does not have to be after the end of PS interviews
- Aims & Objectives
    - Current one is too technical
    - add explanation for scope, users, existing solutions, current problems
    - Mission and what you plan to do
- Requirements
    - what users / stakeholders need
    - interviews and surveys
    - findings from interviews
- Assumption testing
    - methodology and steps from interview
    - how the interviews were conducted
- Planning
    - Plan for 9 weeks (let's say duration of timeline is 30th June – 2nd September 2024?)

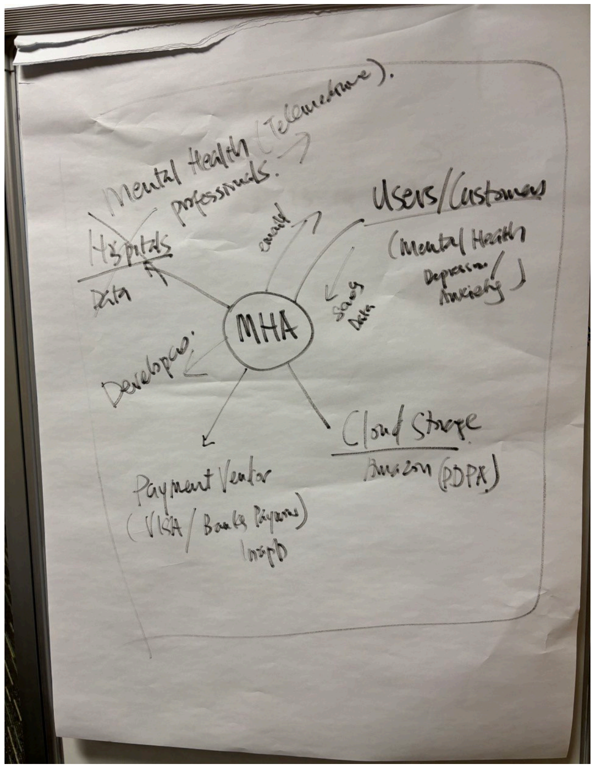
Action items (Main Deadline: 30th June)

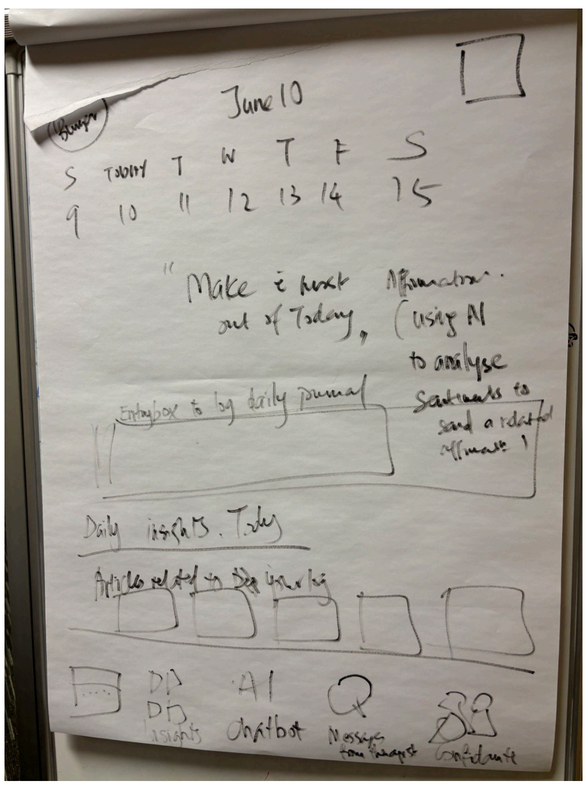
- [x] ~~URGENT: Individual tasks and sections to be started and completed~~
- [x] ~~Start with interviews to determine the problem statement (one teammate interview one)~~

- [x] ~~Make changes to sections in the proposal according to the lecturer's advice (refer to recording)~~
    - [x] ~~Aims and Objectives~~
    - [x] ~~Scope~~
    - [x] ~~Requirements + Stakeholders~~
    - [x] ~~Assumption Testing~~
- [x] ~~Compile survey results~~
- [x] ~~Wireframe for app (+ initial prototypes)~~
- [x] ~~Devon: Survey and interview size for Requirements elicitation~~
    - [x] ~~Refer to audio recording and picture for lecturer's advice~~

<u>Initial planning</u>




Mental Health (Telemedicine).
professionals.
~~Hospitals~~
Data
email
Users/Customers
(Mental Health
Depression/
Anxiety)
MHA
Developer
Cloud Storage
Amazon (PDPA)
Payment Vendor
(VISA / Banks Payment)


June 10
S TODAY T W T F S
9 10 11 12 13 14 15
"Make the most out of Today"
(using AI to analyse sentiments to send a related affirmation)
Entry box to log daily journal
Daily insights. Today
Insights
Chatbot
Message from therapist
Confidante